

2025-12-05 - Valentini (17)

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Quasi linear theory

We begin from the Vlasov-Poisson system

$$\begin{cases} \frac{\partial f(x, v, t)}{\partial t} + v \frac{\partial f(x, v, t)}{\partial x} - \frac{eE(x, t)}{m_e} \frac{\partial f(x, v, t)}{\partial v} = 0 \\ \frac{\partial E(x, t)}{\partial x} = 4\pi e \left[n_0 - \int_{-\infty}^{\infty} f(x, v, t) dv \right] \end{cases}$$

Where we define the distribution function and the electric field as

$$f(x, v, t) = f_0(v) + f_1(x, v, t), \quad \text{with } f_1(x, v, t) = \sum_k f_k(v, t) e^{ikx}$$

$$E(x, t) = E_1(x, t), \quad \text{with } E_1(x, t) = \sum_k E_k(t) e^{ikx}$$

we also recall that

$$\langle f_1(x, v, t) \rangle = \langle E_1(x, t) \rangle = 0$$

with the averages carried out in the physical space

$$\langle \dots \rangle = \frac{1}{L} \int_0^L (\dots) dx$$

We want to allow the velocity distribution to be disturbed in the velocity space significantly.

We consider $f(v, t)$, so we make it vary in time

$$\begin{aligned} f(x, v, t) &= \cancel{f_0(v) + f_1(x, v, t)} \\ &= f_0(v, t) + f_1(x, v, t) \end{aligned}$$

so, as a consequence, we get

$$\langle f_1(x, v, t) \rangle = f_0(v, t)$$

Now we take the average value of the Vlasov equation

$$\begin{aligned} \frac{\partial}{\partial t} \langle f \rangle + v \left\langle \cancel{\frac{\partial f}{\partial x}} \right\rangle - \frac{e}{m} \left\langle \cancel{E_1 \frac{\partial f_0}{\partial v}} \right\rangle - \frac{e}{m} \frac{\partial}{\partial v} \langle E_1 f_1 \rangle &= 0 \\ \implies \frac{\partial}{\partial t} \langle f \rangle &= \frac{e}{m} \frac{\partial}{\partial v} \langle E_1 f_1 \rangle \end{aligned}$$

We consider the following term

$$\langle E_1 f_1 \rangle = \frac{1}{L} \int_0^L \sum_{k'} \sum_k E_{k'}(t) f_k(v, t) e^{i(k'-k)x} dx$$

which is different from zero only for $k' = k$, so

$$\begin{aligned}\langle E_1 f_1 \rangle &= \sum_k E_{-k}(t) f_k(v, t) \\ \Rightarrow \frac{\partial}{\partial t} \langle f \rangle &= \frac{e}{m} \frac{\partial}{\partial v} \sum_k E_{-k}(t) f_k(v, t)\end{aligned}$$

Now we use the linear Landau result to obtain a first perturbative nonlinear result.

$$E_1(x, t) = \sum_k E_k(t) e^{ikx} = \sum_k E_k(0) e^{-i\omega_k t} e^{ikx}$$

Later we will use the following notation

$$\omega_k = \omega_{R,k} + \omega_{I,k}$$

We will use the Green's function.

We move to the Fourier space

$$\frac{\partial}{\partial t} f_k(v, t) + ikv f_k(v, t) - \frac{e}{m} E_k(t) \frac{\partial f_0}{\partial v} = 0$$

As exercise:

Verify that

$$f_k(v, t) = \int_0^t dt' \frac{e}{m_e} E_k(t') \frac{\partial f_0}{\partial v} e^{ikv(t'-t)}$$

Green function

if we plug in a cosine, we obtain a previous solution.

To solve this, first we solve the integral and then we plug the result in Vlasov equation in order to get $0 = 0$.

So, we solve the integral

$$\begin{aligned}f_k(v, t) &= \int_0^t dt' \frac{e}{m_e} E_k(0) e^{-i\omega_k t'} \frac{\partial f_0}{\partial v} e^{ikv t'} e^{-ikv t} = \\ &= \frac{e}{m_e} E_k(0) \frac{\partial f_0}{\partial v} e^{-ikv t} \int_0^t dt' e^{-i\omega_k t'} e^{ikv t'} = \\ &= \frac{e}{m_e} E_k(0) \frac{\partial f_0}{\partial v} e^{-ikv t} \int_0^t dt' e^{i(kv - \omega_k) t'} = \\ &= \frac{e}{m_e} E_k(0) \frac{\partial f_0}{\partial v} e^{-ikv t} \frac{e^{i(kv - \omega_k) t} - 1}{i(kv - \omega_k) t}\end{aligned}$$

If we plug this into Vlasov equation, we obtain $0 = 0$

$$\begin{aligned}f_k(v, t) &= \frac{e}{m_e} \frac{\partial f_0}{\partial v} E_k(0) (e^{-i\omega_k t} e^{i\omega_k t}) e^{-ikv t} \frac{e^{i(kv - \omega_k) t} - 1}{i(kv - \omega_k) t} = \\ &= \frac{e}{m_e} \frac{\partial f_0}{\partial v} E_k(t) \frac{1 - e^{-i(kv - \omega_k) t}}{i(kv - \omega_k) t}\end{aligned}$$

where in the second passage we used

$$E_k(t) = E_k(0)e^{-i\omega_k t}$$

We can identify in this expression the Resonant function in the general case (before we had a look at a particular case of Resonant function)

$$\begin{aligned}\mathbb{R} &\equiv \frac{1 - e^{-i(kv - \omega_k)t}}{i(kv - \omega_k)t} \\ \frac{\partial}{\partial t} \langle f \rangle &= \frac{e}{m} \frac{\partial}{\partial v} \sum_k |E_k(t)|^2 \frac{\partial f_0}{\partial v} \mathbb{R} = \\ &= \frac{\partial}{\partial v} \left[\frac{e^2}{m^2} \sum_k |E_k(t)|^2 \frac{\partial f_0}{\partial v} \mathbb{R} \right] = \\ &= \frac{\partial}{\partial t} f_0\end{aligned}$$

Renormalization theory

We know that we can approximate the Dirac delta using the following formulation

$$\begin{aligned}\mathbb{R} &= \frac{\Delta\omega}{(kv - \omega_k)^2 + \Delta\omega^2} \longrightarrow \delta \\ \frac{\partial f_0}{\partial t} &= \frac{\partial}{\partial v} \left[\frac{e^2}{m^2} \sum_k |E_k(t)|^2 \frac{\Delta\omega}{(kv - \omega_k)^2 + \Delta\omega^2} \frac{\partial f_0}{\partial v} \right]\end{aligned}$$

This formula tells us how f_0 evolves in time. This time dependency of f_0 happens mostly in the resonant region (because at the denominator we have $kv - \omega_k$).

We approximate this result close to v_ϕ

$$\frac{\partial}{\partial t} f(v \approx v_\phi, t) = \frac{\partial}{\partial v} \left[\frac{e^2}{m^2} \sum_k |E_k(t)|^2 \frac{1}{\Delta\omega} \frac{\partial f_0}{\partial v} \right]$$

and we can rewrite this equation as

$$\frac{\partial f_0}{\partial t} - \frac{e^2}{m^2} \frac{\sum_k |E_k(t)|^2}{\Delta\omega} \frac{\partial^2 f_0}{\partial v^2} = 0 \quad \implies \quad \frac{\partial f_0}{\partial t} - D \frac{\partial^2 f_0}{\partial v^2} = 0$$

This is the velocity diffusion equation (similar to the heat equation in velocity space).

The heat equation tends to make gradients smaller

We can see that it's not really a plateau.

What is the characteristic time of this diffusion?

We use dimensional analysis ($v^2 = \omega^2/k^2$)

$$\begin{aligned}
\frac{f_0}{t} &\sim \frac{e^2}{m^2} E^2 \frac{f_0}{v^2} t = \frac{e^2}{m^2} E^2 k^2 t^3 f_0 \\
\Rightarrow \frac{e^2}{m^2} E^2 k^2 t^4 &\sim 1 \\
\Rightarrow t^2 &\sim \frac{m}{eEk} \\
\Rightarrow t &\sim \sqrt{\frac{m}{eEk}} \rightarrow \tau_t
\end{aligned}$$

with τ_t trapping time.

Last step to full nonlinear model

Nonlinear wave-particle interaction

Physical Review 168, 1957, Berratin and GLK

For $\alpha = i, e$ we have

$$\begin{cases} \frac{\partial f_\alpha}{\partial t} + v \frac{\partial f_\alpha}{\partial x} - \frac{q_\alpha}{m_\alpha} \frac{\partial \phi}{\partial x} \frac{\partial f_\alpha}{\partial v} = 0 \\ \frac{\partial^2 \phi}{\partial x^2} = -4\pi \left[\sum_\alpha q_\alpha \int_{-\infty}^{\infty} f_\alpha(x, v, t) dv \right] = -4\pi \sum_\alpha n_\alpha(x, t) \end{cases}$$

We want stationary condition

$$\frac{\partial}{\partial t} = 0$$

so

$$v \frac{\partial f_\alpha}{\partial x} - \frac{q_\alpha}{m_\alpha} \frac{\partial \phi}{\partial x} \frac{\partial f_\alpha}{\partial v} = 0$$

and $f_\alpha(x, v, t) = f_\alpha(x, v)$

Trapped particles equilibria

We say that ξ is a constant of motion and we say that

$$f_\alpha = f_\alpha(\xi)$$

is a solution of the Vlasov equation. (we saw this at the beginning of the course)

$$\xi = \frac{1}{2} m v^2 + U(X) = \begin{cases} \frac{1}{2} m_i v^2 + e\phi(x) & \text{for ions} \\ \frac{1}{2} m_e v^2 - e\phi(x) & \text{for electrons} \end{cases}$$

On one hand, for ions we have

Given an energy E , the electric potential hole is depicted above, while the potential energy hole is drawn below.

Ions are trapped close to the minima of the electric potential.

$$\mathcal{E} > U_{max} \quad \text{free ions}$$

$$U_{min} < \mathcal{E} < U_{max} \quad \text{free electrons}$$

$$e\phi_{min} < \mathcal{E} < e\phi_{max}$$

On the other hand, for electrons we have

Given an energy E , electrons are trapped close to the minimum of the potential energy, which corresponds to the maximum of the electric potential.

$$\mathcal{E} > U_{max} = -e\phi_{min} \quad \text{free electrons}$$

$$U_{min} < \mathcal{E} < U_{max} \quad \text{trapped electrons}$$

$$-e\phi_{max} < \mathcal{E} < -e\phi_{min}$$

So we have to be careful because ions are trapped at the minimum of ϕ , while electrons are trapped at the maximum of ϕ .

$$\begin{aligned} \frac{d^2\phi}{dx^2} &= 4\pi e \left[\int_{-\infty}^{\infty} f_e dv - \int_{-\infty}^{\infty} f_i dv \right] = \\ &= 4\pi e \left[\int_{-\infty}^0 f_e dv + \int_0^{\infty} f_e dv - \int_{-\infty}^0 f_i dv - \int_0^{\infty} f_i dv \right] \end{aligned}$$

In order for our hypothesis to hold, we need a null magnetic field, so we need f_i and f_e to be symmetric with respect to the origin.

$$\frac{d^2\phi}{dx^2} = 4\pi e \left[2 \int_0^{\infty} f_e dv - 2 \int_0^{\infty} f_i dv \right]$$

We apply a change of variable

$$\mathcal{E} = \frac{1}{2} m_{\alpha} v^2 + q_{\alpha} \phi \implies v = \pm \sqrt{2 \left[\frac{\mathcal{E}}{m} + \frac{q_{\alpha}}{m_{\alpha}} \phi \right]}$$

and also

$$dv = \pm \frac{2/m_{\alpha}}{2\sqrt{\frac{2}{m_{\alpha}}[\mathcal{E} - q_{\alpha}\phi]}} d\mathcal{E} = \pm \frac{d\mathcal{E}}{\sqrt{2m_{\alpha}[\mathcal{E} - q_{\alpha}\phi]}}$$

and

$$\frac{d^2\phi}{dx^2} = 4\pi e \left[2 \int_{-e\phi}^{\infty} \frac{f_e(\xi) d\xi}{\sqrt{2m_e[\mathcal{E} + e\phi]}} - 2 \int_{e\phi}^{\infty} \frac{f_i(\xi) d\xi}{\sqrt{2m_i[\mathcal{E} - e\phi]}} dv \right]$$

We define

- $f_{\text{TR}}(\xi)$ as the distribution function of trapped electrons
 - $f_{\text{UN}}(\xi)$ as the distribution function of untrapped electrons
- they satisfy

$$f_e(\xi) = f_{\text{UN}}(\xi) + f_{\text{TR}}(\xi)$$

We know

$$\phi, \quad f_i(\xi), \quad f_{\text{UN}}(\xi)$$

but we don't know $f_{\text{TR}}(\xi)$.

We can rewrite Poisson equation as

$$\begin{aligned} \frac{d^2\phi}{dx^2} = 4\pi e \left[2 \int_{-e\phi}^{-e\phi_{\min}} \frac{f_{\text{TR}}(\xi) d\xi}{\sqrt{2m_e [\mathcal{E} + e\phi]}} + \right. \\ \left. + 2 \int_{-e\phi_{\min}}^{\infty} \frac{f_{\text{UN}}(\xi) d\xi}{\sqrt{2m_e [\mathcal{E} + e\phi]}} + \right. \\ \left. - 2 \int_{e\phi}^{\infty} \frac{f_i(\xi) d\xi}{\sqrt{2m_i [\mathcal{E} - e\phi]}} dv \right] \end{aligned}$$

Now we isolate on the left the integral containing $f_{\text{TR}}(\xi)$

$$2 \int_{-V}^{-V_{\min}} \frac{f_{\text{TR}}(\xi) d\xi}{\sqrt{2m_e [\mathcal{E} + V]}} = \underbrace{\frac{1}{4\pi e} \frac{d^2}{dx^2} V - \int_{f_{\text{UN}}} (\dots) + \int_{f_i} (\dots)}_{\text{known}}$$

the right hand side is difficult, but eventually we can evaluate the term.

"This problem is an integral equation of convolution type which is easy to solve using a standard Laplace transform." said the article above

The convolution is defined as follow

$$f * g \equiv \int_0^t h(x)g(t-x) dx, \quad \begin{cases} h(x) = 0 & \Longleftrightarrow x < 0 \\ g(t-x) = 0 & \Longleftrightarrow t-x < 0 \end{cases}$$

And its Laplace transform is

$$\mathcal{L}[f * g] = \mathcal{L}[h(t)](p) \mathcal{L}[g(t)](p)$$

We solve the integral equation above using a change of variable

$$x = \xi + v \quad \Longrightarrow \quad \xi = x - v \quad \Longrightarrow \quad d\xi = dx$$

So the left hand side of the equation becomes

$$\begin{aligned} \int \frac{2f(x-v) dx}{\sqrt{2m_x}} \\ v - v_{\min} = t \quad \Longrightarrow \quad v = t + v_{\min} \\ \int_0^t \frac{2f(x-t-v_{\min}) dx}{\sqrt{2m_x}} = W(t+v_{\min}) = W(t) \end{aligned}$$

Now we can identify

$$\begin{cases} h(x) = \sqrt{\frac{2}{m}} \frac{1}{\sqrt{x}}, & \tilde{h}(x) = \frac{1}{\sqrt{x}} \\ g(t-x) = f(x-t-v_{\min}) \end{cases}$$

which let us write the integral as

$$\int_0^t g(t-x)h(x) dx = W(t)$$

Now we calculate the Laplace transform of both terms

$$\mathcal{L}[h * g](p) = \mathcal{L}[W(t)](p)$$

with

$$\mathcal{L}[g(t)](p) = \frac{\mathcal{L}[W(t)](p)}{\mathcal{L}[h(t)](p)} \cdot \frac{p}{p} = \frac{\mathcal{L}[W'(t)](p)}{p \cdot \mathcal{L}[h(t)](p)}$$

where we used

$$\mathcal{L}[W'(t)](p) = p \cdot \mathcal{L}[W(t)](p) - \underbrace{W(t > 0)}_0$$

(we assumed the last term to be zero)

We go on

$$\begin{aligned} \mathcal{L}[g(t)](p) &= \frac{\mathcal{L}[W'(t)](p)}{p\sqrt{2/m} \cdot \mathcal{L}[\tilde{h}(t)](p)} = \\ &= \frac{\mathcal{L}[W'(t)](p)}{p\sqrt{2/m} \cdot \sqrt{\pi/p}} = \\ &= \frac{\mathcal{L}[W'(t)](p)}{p\sqrt{2/m} \cdot \sqrt{\pi/p} \cdot \sqrt{\pi/\pi}} = \\ &= \sqrt{\frac{m}{2}} \mathcal{L}[W'(t)](p) \frac{1}{\pi} \sqrt{\frac{\pi}{p}} = \\ &= \sqrt{\frac{m}{2}} \mathcal{L}[W'(t)](p) \frac{1}{\pi} \mathcal{L}[\tilde{h}(t)](p) = \\ &= \sqrt{\frac{m}{2}} \frac{1}{\pi} \mathcal{L}\left[\int_0^t \tilde{h}(t-V)W'(V) dV\right](p) \end{aligned}$$

So, substitung back to the original variables

$$\begin{aligned} \mathcal{L}[g(t)](p) &= \sqrt{\frac{m}{2}} \frac{1}{\pi} \mathcal{L}\left[\int_0^t \frac{1}{\sqrt{t-V}} \frac{dW(V)}{dV} dV\right] \\ \implies g(t) &= \sqrt{\frac{m}{2}} \frac{1}{\pi} \int_0^t \frac{1}{\sqrt{t-V}} \frac{dW(V)}{dV} dV \\ \implies g(t-x) &= \sqrt{\frac{m}{2}} \frac{1}{\pi} \int_0^{t-x} \frac{1}{\sqrt{t-x-V}} \frac{dW(V)}{dV} dV \\ \implies f(x-t-v_{min}) &= \sqrt{\frac{m}{2}} \frac{1}{\pi} \int_0^{t-x} \frac{1}{\sqrt{t-x-V}} \frac{dW(V)}{dV} dV \end{aligned}$$

But we know that

$$x-t-v_{min} = (\mathcal{E} + V) - (V - V_{min}) - V_{min} = \mathcal{E}$$

$$x - t = \mathcal{E} + V_{min} \quad \Longrightarrow \quad t - x = -(\mathcal{E} + V_{min})$$

So

$$f(\xi) = \sqrt{\frac{m}{2}} \frac{1}{\pi} \int_0^{-\mathcal{E} - V_{min}} \frac{1}{\sqrt{-\mathcal{E} - V_{min} - V}} \frac{dW(V)}{dV} dV$$

We change variables again

$$V' = V + V_{min} \quad \Longrightarrow \quad dV = dV'$$

We recall the definition of convolution

$$f * g \equiv \int_0^t h(x) g(t - x) dx, \quad \begin{cases} h(x) = 0 & \Longleftrightarrow x < 0 \\ g(t - x) = 0 & \Longleftrightarrow t - x < 0 \end{cases}$$

and its more generic definition

$$f * g \equiv \int_{-\infty}^{\infty} h(x) g(t - x) dx$$

So

$$\begin{aligned} f(\xi) &= \sqrt{\frac{m}{2}} \frac{1}{\pi} \int_{V_{min}}^{-\mathcal{E}} \frac{1}{\sqrt{-\mathcal{E} - V'}} \frac{dW(V')}{dV'} dV' = \\ &= \sqrt{\frac{m}{2}} \frac{1}{\pi} \int_{e\phi_{min}}^{-\mathcal{E}} \frac{1}{\sqrt{-\mathcal{E} - V'}} \frac{dW(V')}{dV'} dV' \end{aligned}$$

Also

We assumed

$$W(t = 0) = 0 \quad \Longrightarrow \quad W(\underbrace{V - V_{min}}_{=0}) = 0 \quad \Longrightarrow \quad W(V_{min}) = 0$$

We have

$$h(x) = \sqrt{\frac{2}{m}} \frac{1}{\sqrt{x}}$$

and to keep it real we assume

$$h(x) = 0, \quad x < 0$$

otherwise it's imaginary.

At last

$$\begin{aligned} g(t - x) &= f(x - t - V_{min}) \\ t - x < 0 &\Longrightarrow V - V_{min} - \mathcal{E} - V < 0 \\ &\Longrightarrow \mathcal{E} > -V_{min} \\ &\Longrightarrow \mathcal{E} > -e\phi_{min} \end{aligned}$$

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